

STANDARD PRACTICE

~ 10^4 FEM simulations → fit a surrogate
→ optimise through its approximation error

THIS WORK

0 stored simulations → differentiate the physics itself → exact gradients, no surrogate

DEVICE UNDER DESIGN

Electrode gap
profile $d(x)$

Travel spec T ,
boundary condition

solve

gradient

DIFFERENTIABLE PULL-IN SOLVER

pull-in is a saddle-node bifurcation – we solve for it directly

$$R(Y, \Lambda) = 0 \quad J(Y, \Lambda) v = 0 \quad v^T v - 1 = 0$$

extended fold system (Keller 1977)
returns V_{PI} , stable travel, deflection mode v

exact $\partial V_{PI}/\partial d(x)$ by the implicit function theorem – differentiated through the instability

1 INVERSE DESIGN

Gradient descent on the
free-form gap profile,
every iterate feasible

**22.86 V → 13.60 V (–41%)
geometry only, in 15 s**

2 AMORTIZED NETWORK

Maps spec → geometry,
trained end-to-end
through the solver

**0 stored simulations
0.04 ms per design**

3 SAFE-OPERATION RL

Infers each device's own
fabrication ceiling,
which is not measurable

**+12.8% usable travel
0.0% of devices destroyed**

VALIDATED AGAINST 8 INDEPENDENT SOURCES, 1967-2025

exact gradient matches the analytic c^3 law to 0.00% • optimal-exponent reversal between cantilever and fixed-fixed beams reproduced